

Efficient Intrusion Detection for Edge Network via Multi-Stage Few-Shot Class-Incremental Learning

Zhen Yin , Wei Wang , *Member, IEEE*, Yanyu Cheng , *Senior Member, IEEE*, Yiliang Liu , *Member, IEEE*, and Xiaozhen Lu , *Member, IEEE*

Abstract—The emerging increasingly sophisticated, intelligent, and stealthy network attacks pose more severe security threats to the edge network. In particular, the emergence of novel intelligent attacks makes it a challenging issue to obtain sufficient attack samples, and thus classical deep learning-driven intrusion detection frameworks (IDSs) become ineffective. To tackle this issue, we introduce a novel intrusion detection framework leveraging few-shot class-incremental learning (FSCIL) capabilities to achieve robust detection of emerging threats with few samples. This approach pre-trains a backbone traffic classification model and employs few-shot training with prototypical networks. To further reduce catastrophic forgetting while improving both accuracy and system robustness, we incrementally fine-tune the classification model with supervised contrastive learning, and also realize rapid adaptation to new attacks. Evaluations on the intrusion detection datasets CIC-IDS2017 and USTC-TF2016 demonstrate that our framework consistently outperforms baseline models for emerging attacks detection with few attack samples while preserving effective recognition of known threats.

Index Terms—Network intrusion detection, few-shot class-incremental learning (FSCIL), edge network, deep learning.

I. INTRODUCTION

THE sixth-generation (6G) wireless network will evolve toward open and diversified services, heterogeneous and cooperative resources, and enhanced trust and security, with the support of super edge nodes [1]. However, the collaboration among heterogeneous edge nodes belonging to different network

operators inevitably introduces trust issues, which further intensified by the massive access of untrusted devices [2]. To support secure and trusted network services, continuous edge nodes monitoring is in urgent demand. Moreover, emerging attacks are becoming increasingly diverse and unpredictable, posing significant challenges for real-time attack detection owing to insufficient timely and sufficient labeled new attack samples.

Network intrusion detection system (IDS) provides an effective solution for real-time attack detection and monitoring for edge networks, serving as a crucial security mechanism for wireless edge networks by isolating intrusions before they cause serious damage [3]. IDS analyzes network traffic and detect malicious attack patterns to distinguish them from legitimate flows. Authors in [4] first proposed the intrusion detection system model in 1987, by introducing a baseline host-based detection strategy with security standards to identify deviations from normal behaviors. Early IDS primarily depends on port-based detection [5] or deep packet inspection (DPI) [6] techniques. Port-based detection may suffer from port masquerading and dynamic port allocation, while DPI is incapable to deal with encrypted traffic data.

Recent machine learning advancements have enhanced classification capabilities, algorithms such as support vector machines (SVM) [7] and k-nearest neighbors (KNN) [8] have been utilized to effectively categorize network traffic patterns. However, these methods rely on manually extracted statistical or time-series features, and thus cannot well adapt to complex and dynamic environments. Whereas deep learning automatically learns hierarchical feature representations directly from raw data, which eliminates manual feature engineering for classification [9]. Wang et al. [10] presented a CNN-driven detection method that converts raw traffic data into images for end-to-end malicious traffic identification. Similarly, Imrana et al. [11] introduced a bidirectional LSTM framework for intrusion detection that captures temporal dependencies in network traffic sequences.

Note that DL-based traffic detection usually requires a large amount of data. However, in practice, novel attacks emerge rapidly, which makes collecting and labeling sufficient network traffic data for network training a challenging or even impossible task. Deep learning-based IDS with limited samples becomes a critical but challenging issue. To address this issue, researchers have explored few-shot learning approaches for novel attack detection. Xu et al. [12] first introduced a meta-learning framework for few-shot network intrusion detection, demonstrating the potential of meta-learning for rapidly adapting to new

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Zhen Yin is with the College of Electronic and Information Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China (e-mail: yinzen@nuaa.edu.cn).

Wei Wang is with the School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: w25wang@xjtu.edu.cn).

Yanyu Cheng is with the Joint NTU–UBC Research Centre of Excellence in Active Living for the Elderly (LILY), Nanyang Technological University, Singapore 639798 (e-mail: ycheng022@e.ntu.edu.sg).

Yiliang Liu is with the School of Cyber Science and Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: liuyiliang@xjtu.edu.cn).

Xiaozhen Lu is with the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, China (e-mail: luxiaozhen@nuaa.edu.cn).

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attack types with limited samples. Rong et al. [13] proposed an intrusion detection system based on prototypical networks for few-shot intrusion detection. Lu et al. [14] proposed a few shot based model-agnostic meta-learning approach for intrusion detection in IoT security. While these few-shot learning methods show promising results in learning novel attack patterns from scarce data, they face critical limitations when applied to continuously evolving intrusion detection scenarios. Specifically, these methods suffer from catastrophic forgetting when incrementally learning new attack classes, as adapting to new tasks requires updating shared model parameters, which inevitably overwrites previously learned knowledge. In addition, they encounter feature space confusion, where novel and known attack embeddings overlap, resulting in ambiguous decision boundaries.

Few-shot class-incremental learning (FSCIL) addresses these limitations by learning new attack patterns from limited samples while preserving known attack knowledge. Tao et al. [15] first introduced FSCIL for computer vision. Wang et al. [16] utilized siamese and prototypical networks within a meta-learning framework, but it suffers limited representation learning for complex traffic patterns due to reliance on handcrafted statistical features. Du et al. [17] presented an FSCIL framework for IDS that includes a fixed vision transformer (ViT) feature extractor to prevent catastrophic forgetting and a branch classifier module with reprojection and fusion to enhance feature representation across sessions. Di Monda et al. [18] systematically compared few-shot and FSCIL methods for IDS and adopted the rethinking few-shot (RFS) approach.

Nevertheless, applying few-shot class-incremental learning to intrusion detection in 6G edge networks encounters considerable challenges. The rapidly evolving attack patterns render training samples for new attack classes extremely scarce, causing severe class imbalance for effective learning of novel attacks. In addition, feature overlap between new and existing attack classes further increases misclassification rates and degrades detection accuracy.

To tackle these challenges, we propose a multi-stage collaborative intrusion detection framework based on FSCIL. Our approach first pre-trains a lightweight CNN on abundant base-class samples to establish robust feature extraction. Subsequently, we adopt prototype networks for few-shot learning, where CosFace loss expands inter-class margins for novel attack accommodation. During incremental learning, supervised contrastive learning (SupCon) combines novel samples with selected old-class exemplars to consolidate representations while preventing catastrophic forgetting via fine-tuning partial layers. When novel attacks emerge, our framework enables online updates with minimal labeled samples and training epochs, achieving rapid detection of emerging threats while maintaining high accuracy on historical patterns without full retraining. The main contributions are summarized as follows:

- We propose a traffic intrusion detection scheme for complex edge networks based on few-shot class-incremental learning. Our model is pre-trained on a diverse attack dataset and rapidly adapts to novel attacks through minimal-epoch fine-tuning with few labeled samples.

- To address feature overlap in incremental learning, we introduce a forward-backward compatibility mechanism. The forward phase employs CosFace loss during few-shot training to reserve representational margin for novel attacks, and the backward phase uses supervised contrastive learning during incremental updates to maintain old-new class separability and mitigate forgetting.
- To mitigate catastrophic forgetting in incremental learning, we freeze the backbone to preserve pretrained features, fine-tuning only a subset of layers to accommodate new classes. Based on herding, we select representative exemplars and replay them during incremental training, which reinforces old-class knowledge using minimal memory without full retraining.
- Simulation results on the CIC-IDS2017 and USTC-TFC2016 datasets validate that our method achieves exceptional detection performance while significantly mitigating catastrophic forgetting and achieving rapid convergence on novel attacks with only a few labeled samples.

The remainder of this paper is organized as follows. Section II discusses related work. In Session III, we introduce the system model and data preprocessing. In Session IV, we present the proposed few-shot incremental learning scheme for intrusion detection. Simulations and discussions are given in Section V, and conclusions are given in Section VI.

II. RELATED WORK

This section begins with an overview of security attacks in edge networks, followed by a review of related work on intrusion detection and few-shot class-incremental learning approaches.

A. Security Attacks in Edge Network

6G networks target Tbps-scale data rates, Kbps/Hz spectral efficiency, and μ s-scale latency to support novel applications [1], e.g., holographic communication [19] and digital twins [20]. Massive data from these applications overburden the inefficient and high-latency model of traditional cloud computing, driving the deployment of numerous servers at the network edge [21]. Diverse, heterogeneous, and untrusted devices accessing open 6G edge networks expand the attack surface, increasing vulnerability to denial of service (DoS) attacks, zero-day attacks, and similar threats, thereby intensifying cybersecurity risks [22].

IoT networks serve as the most critical application scenario for edge computing in 6G edge networks, yet the computational and communication constraints of IoT devices hinder effective security deployment [23]. Given the massive scale of IoT deployments, data transmission through open wireless edge networks becomes vulnerable to security attacks, particularly forgery, man-in-the-middle, and replay attacks that compromise data reliability [24]. Meanwhile, these vulnerable devices can be coordinated by attackers to launch DDoS attacks that overwhelm edge servers through excessive service requests, while simultaneously targeting IoT devices themselves to cause device failures and service outages [25]. Additionally, malware poses a significant threat to IoT edge networks, as infected devices

can be hijacked to disrupt remote control functions and compromise service availability [26]. Compromised IoT devices and edge servers may also be exploited for unauthorized computational tasks such as cryptocurrency mining, imposing additional electricity costs on legitimate users. The heterogeneous nature of IoT environments with diverse operating systems makes comprehensive malware protection particularly challenging to implement [27].

To achieve the 6G vision of pervasive intelligence and internet of intelligence, native AI integration at the network architecture level becomes essential due to the inherent requirements of AI for distributed processing, real-time responsiveness, and robust security mechanisms [28]. This evolution naturally drives artificial intelligence (AI) capabilities toward the network edge, enabling edge AI to serve as an innovative technology that integrates sensing, communication, computation, and intelligence to enhance network efficiency and security. However, this integration introduces new vulnerabilities, particularly data poisoning attacks where malicious users inject false data to corrupt AI models during both training and inference phases [29]. During training, poisoned data can degrade model accuracy or misdirect the learning process, while during inference, corrupted inputs cause trained models to produce erroneous outputs, ultimately leading to network performance degradation across edge computing, caching, and intelligence services [30].

The inherent vulnerabilities of edge networks, including their open, distributed architecture and integration of diverse IoT devices and AI systems, create a complex threat landscape with multiple attack vectors [31]. Meanwhile, 6G applications such as remote surgery and critical infrastructure control impose unique and stringent security and privacy requirements that demand ultrahigh reliability [32]. Therefore, comprehensive defense strategies must be established to overcome the disconnect between these security vulnerabilities and stringent safety standards. The IDS serves as a vital component in this security framework, enabling real-time identification of malicious behavior and rapid threat mitigation to maintain the robust performance required for edge services [33].

B. Intrusion Detection

Anderson [34] first systematically defined intrusion threats as unauthorized hostile activities targeting data confidentiality, integrity, or system availability. Denning [4] subsequently proposed the first intrusion detection model, introducing a host-based approach that detects security violations by analyzing audit data for irregular system behaviors. Since then, researchers have conducted extensive research on intrusion detection systems. Based on different data sources [35], intrusion detection systems are typically classified into three main types: host-based systems (HIDS) [36], network-based systems (NIDS) [37], and wireless-based systems (WIDS) [38]. HIDS monitor system logs, file integrity, and local activities to detect malicious behavior on individual hosts, but their protection is limited to the specific machine where they are installed [39]. NIDS are deployed throughout the network infrastructure to analyze traffic for network device security, analyzing various

protocol layers including network, transport, and application protocols to identify suspicious network activities [40]. WIDS monitor wireless communication channels and analyze protocol behaviors to detect malicious access points and wireless-specific attacks, though they primarily focus on wireless layer protocols rather than application layer activities [41].

Despite monitoring different data sources, these IDS types employ similar detection techniques that are broadly categorized [42] into signature-based [43] and anomaly-based approaches [44]. Signature-based detection maintains a database of known attack patterns, each representing the digital fingerprint of a specific malicious activity. When monitoring network traffic or system events, the IDS performs pattern matching by comparing incoming data against these predefined signatures to identify potential threats [45]. For instance, Saelee et al. [46] improved signature-based detection through ensemble methods with boolean weighted evaluation for enhanced accuracy. Kunmar et al. [47] demonstrated the practical application of signature-based detection by implementing Snort IDS with the DARPA dataset to analyze real-time network traffic and detect abnormal transmission patterns. However, this method faces limitations in complex and dynamic network environments, as it can only detect previously identified attacks and remains ineffective against novel threats, zero-day exploits, and polymorphic attacks that deliberately alter their characteristics to avoid signature matching [48].

In contrast, anomaly-based detection builds reference patterns of legitimate network activity and triggers alerts when activities deviate significantly from these patterns, enabling detection of unknown attacks that lack existing signatures [49]. Early anomaly-based detection approaches primarily relied on traditional statistical methods [50] or domain expertise [51] to model normal behavior. Nevertheless, with the increasing complexity of network environments, these conventional approaches became challenging to scale and required extensive manual tuning to establish accurate baselines. With advances in machine learning techniques, researchers began exploring machine learning approaches which contain both supervised and unsupervised methods for anomaly-based intrusion detection. Supervised learning trains classifiers using labeled normal and anomalous network traces to detect and categorize intrusions, with widely adopted methods including decision tree (DT) [52], [53], [54], [55], support vector machine (SVM) [56], [57], [58], [59], [60], K-nearest neighbors (KNN) [61], [62], [63], naive bayes (NB) [64], [65], [66], [67] and random forest (RF) [68], [69], [70]. Uwazie et al. [71] systematically compared RF, KNN, SVM for intrusion detection, evaluating their performances across various standard datasets. Unsupervised learning identifies normal behavior patterns and triggers alerts when network activity deviates beyond predefined thresholds, with effective approaches including K-means clustering [72], [73], hierarchical clustering [74], [75] and density-based spatial clustering of applications with noise (DBSCAN) [76], [77].

However, traditional machine learning methods have limitations in automatically extracting meaningful features from complex network data and may struggle with the high-dimensional nature of modern network traffic [78]. Deep learning (DL)

techniques have emerged as a promising approach, employing deep neural architectures to extract complex patterns directly from raw network traffic without manual feature engineering [79]. Various deep learning architectures have shown remarkable effectiveness in intrusion detection applications [80]. Auto-encoders (AE) [81], [82], [83] are particularly effective for anomaly detection as they create efficient encodings of normal network operations and highlight anomalous activities that differ from baseline behavior. Generative adversarial networks (GAN) [84], [85], [86], [87] are adept at generating synthetic attack samples to augment limited training datasets and improve the robustness of intrusion detection models. Convolutional neural networks (CNN) [88], [89], [90], [91] are proficient at extracting local patterns from spatial features of network traffic, capable of identifying malicious behavioral characteristics hidden in packet sequences. Recurrent neural networks (RNN) [92], [93], [94] and their variants, such as long short-term memory (LSTM) [95], [96], bidirectional long short-term memory (BiLSTM) [97], [98] and gated recurrent unit (GRU) [99], [100] are particularly suitable for handling temporal characteristics of network traffic, capturing the dependencies and patterns of attack behaviors. Furthermore, researchers have also explored hybrid approaches combining different architectures such as CNN-RNN [101], [102], CNN-LSTM [103], [104], CNN-BiLSTM [105], [106] and CNN-GRU [107], [108]. Nevertheless, as network environments become increasingly complex and novel attacks emerge with limited samples, traditional deep learning models struggle to adapt quickly [109]. Furthermore, existing approaches often sacrifice overall system performance and reliability when attempting to handle class imbalance in these dynamic environments [110], motivating the exploration of few-shot incremental learning for intrusion detection.

C. Few-Shot Class-Incremental Learning

Continual learning is a machine learning paradigm designed to enable models to acquire new knowledge incrementally while preserving previously learned information. In real-world applications, the limited availability of samples often renders traditional continual learning methods ineffective, as scarce labeled data during incremental learning sessions typically triggers model overfitting on novel classes and erosion of previous knowledge. Since Tao et al. [15] initially proposed few-shot class-incremental learning for computer vision, many researchers have extended this paradigm to various applications. Existing FSCIL approaches are generally classified into forward-compatible methods, which train on base classes and keep the model frozen during incremental sessions, and incremental fine-tuning methods, which adjust the model in incremental sessions [111].

Forward-compatible methods aim to enhance base class separability in the embedding space, reducing interference from new classes added incrementally. Notably, Peng et al. [112] employ a CosFace penalty strategy to achieve tight within-class grouping and distinct between-class margins, enabling effective integration of new classes within the embedded feature space. Zhou et al. [113] propose Forward Compatible Training (FACT)

for FSCIL, leveraging simulated prototypes to optimize existing class encodings while preserving capacity for future classes. Oh et al. [114] introduce the CLOSER framework, which reduces inter-class distances in FSCIL to enhance the balance between discriminability and transferability.

Incremental fine-tuning methods adjust the encoder during incremental sessions to enhance flexibility for learning new tasks, thereby improving adaptability. Mazumder et al. [115] select non-critical parameters for updates during incremental learning. Zhao et al. [116] selectively update model components at varying rates to effectively balance adaptation to new knowledge with retention of prior knowledge. Kang et al. [117] apply the lottery ticket hypothesis to identify a subnetwork of critical parameters that remains frozen, while other parameters are fine-tuned during incremental sessions. Cheraghian et al. [118] develop a semantically-enhanced knowledge distillation framework [119] that employs word embeddings as linguistic guidance to improve the distillation process and mitigate memory degradation in few-shot incremental scenarios.

D. Few-Shot Class-Incremental Learning for Intrusion Detection

As shown in Table I, we comprehensively review key literature on few-shot learning, continual learning, and few-shot class incremental learning in IDS. Since Xu et al. [12] introduced meta-learning to IDS in 2020, researchers have increasingly focused on advancing the generalization ability of trained intrusion detection models to novel attacks.

Continual Learning enables IDS to rapidly adapt to new attack categories with adequate samples in dynamic threat environments while minimizing catastrophic forgetting of prior knowledge. Data et al. [120] proposed a tree-like deep feedforward neural network (T-DFNN) model that supports incremental learning of new categories via dynamic branch expansion. Jin et al. [123] explored federated incremental learning, allowing IDS models to update efficiently in data-siloed environments to address new threats. Cerasuolo et al. [122] compared various continual learning methods in IDS applications, providing insights for technology selection.

Few-shot learning enables IDS models to learn features from few samples using meta-learning or metric-based methods, well-suited for scenarios with limited attack samples. For instance, [12], [14] utilized model-agnostic meta-learning (MAML), [13], [121] employed metric-based prototype networks, and [124] adopted contrastive triplet deep learning networks. However, few-shot learning in IDS primarily faces catastrophic forgetting, where adapting to new attack categories impairs prior detection performance.

Novel attack samples in cybersecurity practice are typically scarce, necessitating IDS to adapt rapidly to new attack types while preserving accuracy for known categories. Few-shot class incremental learning integrates continual learning and few-shot learning to learn new attack categories from limited samples while retaining prior knowledge, making it an optimal solution for online IDS. Currently, few studies have explored the application of FSCIL to intrusion detection, as discussed below.

TABLE I
COMPARISON OF THE RELATED WORK

References	FSL	CIL	Method	Dataset	Input Data
Xu et al. [12], 2020	✓	×	Meta-Learning Framework + feature extraction network and comparison network (FC-Net)	ISCX2012, CIC-IDS2017	First N packet header byte segments
Wang et al. [16], 2021	✓	✓	Model-Agnostic Meta-Learning + Model Merging	NSL-KDD	Flow based-statistics
Rong et al. [13], 2021	✓	×	Prototypical Networks	MCFP, CICInvesAndMal2019, USTC-TFC, BEN-1, self-built dataset	First N bytes of a biflow
Data et al. [120], 2021	×	✓	Multiple Deep Feedforward Neural Network(T-DFNN)	CICIDS2017	Flow based-statistics
Qiang et al. [121], 2022	✓	×	Cosine Similarity-based Classifier	APIMDS	Malware API Call Sequences
Cerasuolo et al. [122], 2023	×	✓	Fine-Tuning, Fine-Tuning with Memory, Learning Without Forgetting, Incremental Classifier and Representation Learning, Bias Correction	IoT-23	First N packet header statistical features
Jin et al. [123], 2023	×	✓	Federated Incremental Learning	ToN-IoT, BoT-IoT, NSL-KDD	Flow based-statistics
Lu et al. [14], 2023	✓	×	Model-Agnostic Meta-Learning	CICDDoS2019, CIC-IDS2017, CSE-CIC-IDS2018, NSL-KDD, UNSW-NB15	Flow based-statistics
Du et al. [17], 2024	✓	✓	Self-Supervised Pretraining + Reprojection Branch Classifier	CIC-IDS2017, CIC-IDS2018	First N packet header byte segments
Li et al. [124], 2024	✓	×	Triplet Deep Learning Network	Self-Constructed App Traffic Dataset	First N bytes of a flow
Niknami et al. [125], 2024	✓	×	Prototypical Network	CIC-IDS2017, CIC-IDS2018	Flow based-statistics
Di Monda et al. [18], 2024	✓	✓	Matching Networks, Prototypical Networks, MetaOptNet, Fine-Tuning, Freezing, Rethinking Few-Shot	CIC-IDS2018, IoT-NID	First N packet header statistical features
Chai et al. [126], 2025	✓	✓	Variational Autoencoder + Dynamic Evolutionary Classifier	BODMAS, NSFOCUS	Malware binary executables raw bytes
our work	✓	✓	Pretraining + CosFace-based Prototypical Networks + Supervise Contrastive Learning	CIC-IDS2017, USTC-TF2016	First N packet header byte segments

Du et al. [17] applied self-supervised pretraining to generate feature representations and froze the feature network during the projection stage with a reprojection branch classifier to reduce feature overlap between new and old attack categories. Chai et al. [126] utilized a variational autoencoder to generate robust feature representations and employed a dynamic evolutionary classifier to adapt to new malware categories.

Based on the aforementioned review, it is worth noting that, research on few-shot class incremental learning for intrusion detection remains limited. Thus, in this study, we propose a multi-stage few-shot incremental approach, integrating pretraining, forward compatibility, and incremental fine-tuning to enhance IDS performance.

III. SYSTEM MODEL AND DATA PREPROCESSING

A. System Model

We consider an intrusion detection scenario in a wireless edge network environment. The IDS is pre-trained on known attack categories $\mathcal{C}_p$ using a large dataset with sufficient labeled samples. After deployment, the system encounters new attack categories $\mathcal{C}_v$ with only limited labeled samples available.

In this setting, the IDS must rapidly fine-tune to accurately classify novel attacks using few samples from $\mathcal{C}_v$, while retaining knowledge of prior attack categories $\mathcal{C}_p$. Since new attacks emerge continuously, we partition samples of $\mathcal{C}_v$ into

temporal sessions $D = \{D_1, D_2, \dots, D_n\}$ based on arrival time. Each session D_i ($i = 1, \dots, n$) contains K_v -way N_v -shot samples from newly discovered attack categories C_v^i . The training constraint requires that only data from the current session D_i is accessible during training. After completing session i , the IDS must classify all previously learned attack categories: $C_p \cup \bigcup_{j=1}^i C_v^j$.

For implementation, monitoring software deployed across the network infrastructure captures traffic in Pcap format, containing timestamps, source and destination IP addresses, port numbers, protocol types, and payload data for detailed packet-level inspection. The captured traffic undergoes analysis within the IDS. The IDS detects potential intrusion events and automatically executes defensive actions based on predefined policies (e.g., blocking source IP addresses or isolating compromised systems) or issues alerts to administrators for incident response, while logging comprehensive event details for subsequent analysis. When novel attacks emerge with few labeled samples, the IDS rapidly adapts its detection model.

B. Data Preprocessing

Before intrusion detection, raw traffic data requires preprocessing, including cleaning, segmentation, and normalization. We segment traffic into bidirectional flows using the five-tuple (source IP address, destination IP address, source port, destination port, and protocol type) to enable precise modeling of network interactions and characterization of malicious traffic behaviors. Additionally, in traffic segmentation, we employ timeout and periodic reset strategies referencing [17]. The timeout strategy terminates inactive links after 64s, while the periodic reset strategy resets segmentation when flows become excessively large after 120 s.

Following segmentation, feature extraction methods are employed on the segmented traffic data to derive robust representations for intrusion detection. Raw-byte methods extract features from packet headers or payloads, offering detailed protocol representations and detecting subtle anomalies effectively. We select the first M packets from each session and extract the initial L bytes of each packet. Following [127], where $M = 6$ -14 and $L = 100$ -200 are used with abundant training data, we adapt these parameters in our case. Specifically, we follow the effective practices outlined in [12], [17], and set $M = 16$ and $L = 256$. The extracted bytes are padded with zeros if necessary to reach L bytes, with MAC and IP addresses anonymized to mitigate privacy leakage and improve model robustness across varied networks. Each packet's 256 bytes are reshaped into a 4×64 sub-matrix, and the 16 packets are concatenated in temporal order to construct a 64×64 matrix representation, thereby preserving the sequential dependencies within the traffic flow. These bytes are then normalized (divided by 255) to form bidirectional flow feature vectors. This approach captures critical protocol headers (e.g., TCP/UDP fields) and flow patterns from early packets, minimizing computation while supporting early threat detection and enhancing real-time analysis performance.

IV. PROPOSED FEW-SHOT CLASS-INCREMENTAL LEARNING SCHEME FOR INTRUSION DETECTION

This section proposes a few-shot incremental learning framework for edge network intrusion detection, as illustrated in Fig. 1. We introduce a multi-stage collaborative optimization framework that incrementally improves model performance via pre-training, few-shot learning and few-shot class-incremental fine-tuning. By efficiently utilizing limited samples, this method improves adaptability and precision against emerging threats while mitigating catastrophic forgetting to maintain accuracy for established attack types.

A. Pre-Training Stage

In few-shot class-incremental learning for intrusion detection, we address catastrophic forgetting and overfitting in incremental learning via a framework separating the feature extractor and classifier [128]. We pre-train a backbone network on extensive traffic data to yield robust traffic feature representations. During few-shot fine-tuning, we freeze specific parameters to retain prior knowledge and prevent overfitting. Given the limited resources of edge networks, we select a CNN as the backbone network. Specifically, we input the preprocessed bidirectional flow byte features of abundant base class samples into the backbone network. We set $M = 16$ and $L = 256$, so the feature vector of size 4096 ($M \times L$) is reshaped into a 64×64 single-channel input, compatible with the backbone network's convolutional architecture. The detailed structure of the CNN architecture we employ is illustrated in Fig. 2. The network is then pre-trained using the cross-entropy loss function:

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}), \quad (1)$$

where N is the number of samples in the training batch and C is the total number of classes. $y_{i,c} \in \{0, 1\}$ represents the c -th component of the one-hot encoded label for sample i , and $\hat{y}_{i,c}$ denotes the predicted probability of sample i belonging to class c after Softmax normalization. Through rigorous optimization on abundant base class samples, the backbone network achieves robust and discriminative feature embeddings.

B. Few-Shot Training Stage

1) *Few-Shot Tasks*: To strengthen the capability of the backbone network for few-shot learning scenarios, we introduce a few-shot training phase between the pre-training and incremental fine-tuning stages. Using the base class dataset, we construct numerous few-shot tasks in K -way N -shot format, where K -way denotes K classes and N -shot refers to N labeled samples per class. For instance, in a 5-way 5-shot configuration, the model is tasked with classifying 5 classes, each represented by 5 samples. Each task comprises a support set and a query set. The support set contains K classes with N labeled samples per class to facilitate feature extraction, while the query set includes

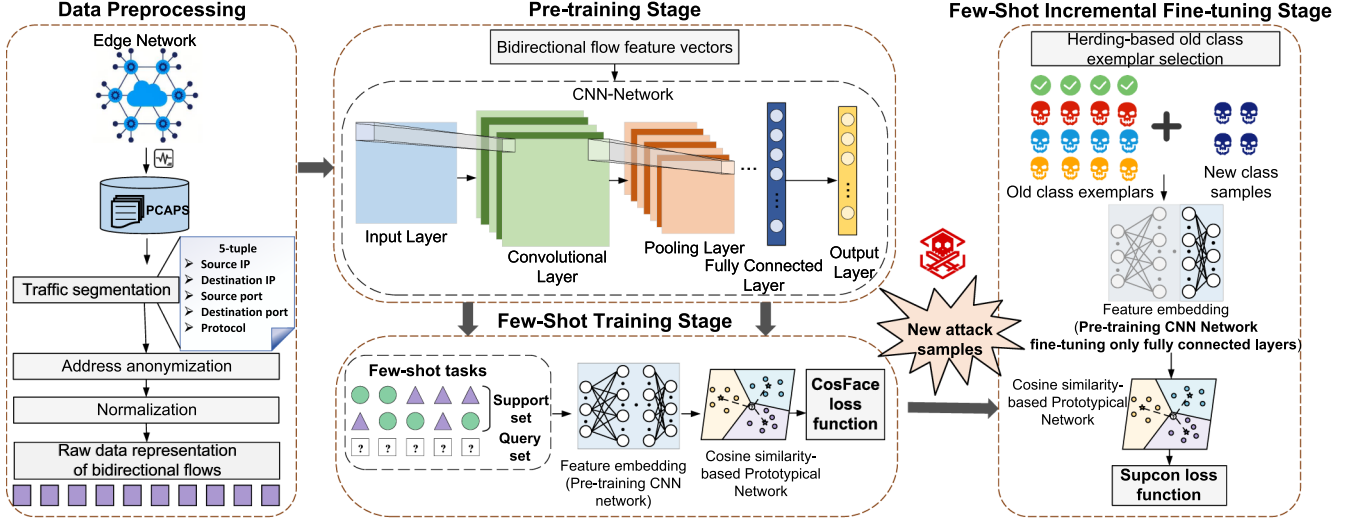


Fig. 1. Workflow of the edge network intrusion detection framework based on Few-Shot Class-Incremental Learning.

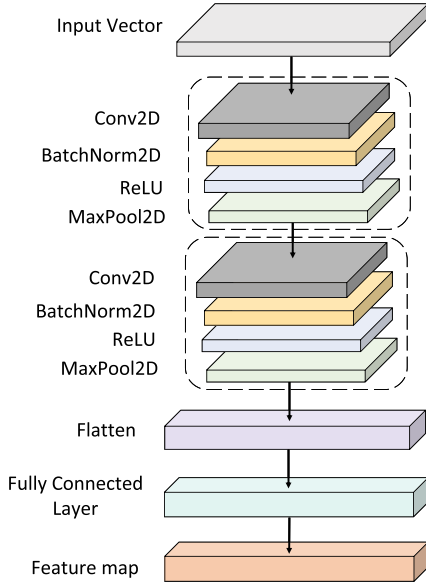


Fig. 2. Architecture of the CNN architecture in this work.

additional unlabeled samples from the same K classes to evaluate the model's classification performance and generalization capability [129].

2) *Prototypical Network*: Features extracted by the backbone network are fed into a prototypical Network [130] for classification. For each of the K classes, the prototypical network computes a prototype c_k by averaging the feature embeddings of the N samples in the support set, and then assigns query samples to the class with the nearest prototype using a distance metric, such as Euclidean distance or cosine similarity, as illustrated in Fig. 3. The prototype c_k for class k is calculated as:

$$c_k = \frac{1}{N} \sum_{(x_i, y_i) \in S_k} f(x_i), \quad (2)$$

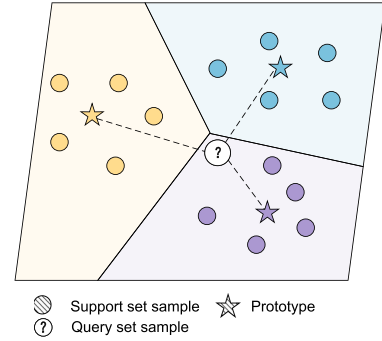


Fig. 3. Schematic of prototypical network for few-shot classification.

where S_k denotes the support set for class k , and $f(x_i) \in \mathbb{R}^d$ is the d -dimensional feature embedding of sample x_i produced by the backbone network. For a query sample x_q with embedding $f(x_q)$, the network computes a distance to each prototype c_k . Here, Cosine similarity acts as the distance metric, defined as:

$$\cos(\theta) = \frac{f(x_q) \cdot c_k}{\|f(x_q)\|_2 \|c_k\|_2}, \quad (3)$$

where $f(x_q) \cdot c_k$ is the dot product, and $\|\cdot\|_2$ denotes the L2 norm.

3) *CosFace Loss Function*: In few-shot incremental learning, the backbone feature extractor must rapidly accommodate novel tasks without compromising prior knowledge. To achieve this, we employ a loss function that contracts within-class variations while expanding cross-class margins, establishing tight categorical groupings and distinct inter-categorical spacing in the representation space. Such organization in the representation space enables efficient accommodation of incrementally arriving classes. Inspired by [112], we adopt the CosFace [131] loss function, defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i})-m)}}{e^{s(\cos(\theta_{y_i})-m)} + \sum_{k \neq y_i} e^{s \cos(\theta_k)}}, \quad (4)$$

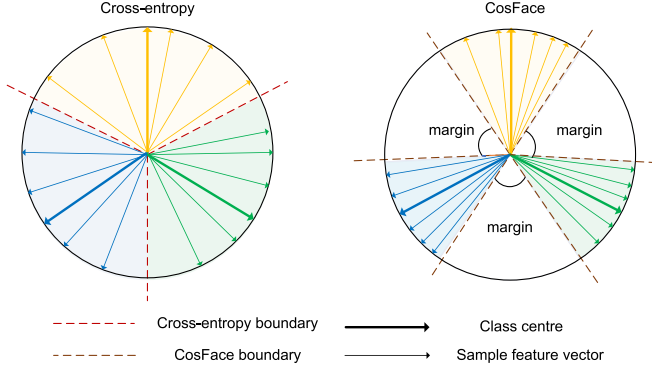


Fig. 4. An illustration of feature distributions: Cross-Entropy loss vs. CosFace loss trained models.

where $\cos(\theta_{y_i})$ represents the cosine similarity between the embedding $f(x_i)$ and the prototype c_{y_i} of its true class y_i , s is a scaling factor, and m is the margin parameter. As shown in Fig. 4, compared to the cross-entropy loss function, by introducing this margin, CosFace enhances inter-class separation and intra-class compactness, thereby improving prototype distinctiveness. This approach, often referred to as forward compatibility in few-shot incremental learning, not only strengthens base class representations for robustness but also preserves flexibility in the feature space, allowing seamless adaptation to new classes without compromising existing ones.

C. Few-Shot Incremental Fine-Tuning Stage

Having utilized the CosFace loss function to promote forward compatibility in few-shot incremental learning, we now enhance model adaptability to novel classes and strengthen model robustness to existing ones through incremental fine-tuning.

1) *Herding-Based Exemplar Selection*: In incremental learning, retaining representative exemplars from old categories is essential to counter catastrophic forgetting and preserve knowledge of prior classes. Selecting highly representative exemplars is crucial for this process, promoting us to adopt the herding strategy introduced by Welling et al. [132] and subsequently improved by Rebuffi et al. [133] for class-incremental learning. Herding uses a greedy approach to form a compact sample subset that effectively reflects the feature distribution of a class. For a class k with a sample set $X_k = \{x_1, x_2, \dots, x_n\}$, where each x_i is a feature embedding output by the backbone network, herding constructs a subset $P_k \subseteq X_k$ of size m , i.e., $P_k = \{p_1, p_2, \dots, p_m\}$ with $|P_k| = m \leq n$, to approximate the true feature centroid of class k . Here, we define the true centroid of class k as:

$$\mu_k = \frac{1}{|X_k|} \sum_{x \in X_k} x, \quad (5)$$

and the subset centroid as:

$$\hat{\mu}_k = \frac{1}{m} \sum_{x \in P_k} x. \quad (6)$$

The method iteratively constructs the subset P_k by selecting one sample at a time, starting from an empty set $P_k = \emptyset$. At each step, it selects a sample x^* that maximizes the cosine similarity between μ_k and the evolving subset mean:

$$x^* = \arg \max_{x \in X_k \setminus P_k} \frac{\mu_k \cdot \hat{\mu}_{k,x}}{\|\mu_k\| \|\hat{\mu}_{k,x}\|_2}. \quad (7)$$

where $\hat{\mu}_{k,x}$ denotes the subset centroid after adding candidate sample x :

$$\hat{\mu}_{k,x} = \frac{1}{|P_k| + 1} \left(\sum_{p \in P_k} p + x \right). \quad (8)$$

For efficient computation, we maintain the cumulative feature sum $S_k = \sum_{p \in P_k} p$ of selected exemplars. At each iteration, for every candidate sample $x \in X_k \setminus P_k$, we compute the cosine similarity between the true class centroid μ_k and the hypothetical subset mean $(S_k + x)/(|P_k| + 1)$, then select the sample maximizing this similarity and update $S_{k+1} = S_k + x^*$. This greedy process continues until $|P_k| = m$, yielding a representative exemplar set that converges faster to μ_k than random sampling and effectively mitigates catastrophic forgetting. These exemplars are then combined with samples from novel classes, including attack samples from incremental sessions D_i ($i = 1, \dots, n$), where each D_i consists of K_{novel} -way N_{novel} -shot samples from the novel class set $\mathcal{C}_{\text{novel}}^i$, during the incremental few-shot fine-tuning stage.

2) *Freezing the Backbone Network for Training*: In the prior data preparation stage, we applied herding to generate exemplars from old class samples, combining these with a limited set of labeled novel class attack samples for incremental fine-tuning. To preserve the generalized feature representation capacity of the backbone network and prevent catastrophic forgetting of prior class knowledge during the integration of novel class samples, we froze the convolutional layers of the CNN, restricting optimization to the fully connected layer. This approach maintains stable representations of existing class features while facilitating precise adaptation to the classification requirements of new class data. Consistent with the few-shot training stage, we also utilize the cosine similarity-based Prototypical Network as the classification framework, which computes class prototypes from support set embeddings to enable efficient and accurate classification of novel classes.

3) *Supervised Contrastive Loss*: During few-shot training, CosFace loss optimizes class boundaries, reserving space for novel classes. Still, as Fig. 5 indicates, new class embeddings may overlap with existing classes, impairing classification accuracy. To address this, in the few-shot incremental phase, we employ supervised contrastive loss (SupCon) [134] to optimize feature representations. SupCon employs contrastive learning to cluster same-class embeddings while separating different-class embeddings, reducing overlap between new and existing classes in the feature space. The SupCon loss is defined as:

$$L_{\text{SupCon}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{e^{z_i \cdot z_p / \tau}}{\sum_{a \in A(i)} e^{z_i \cdot z_a / \tau}}. \quad (9)$$

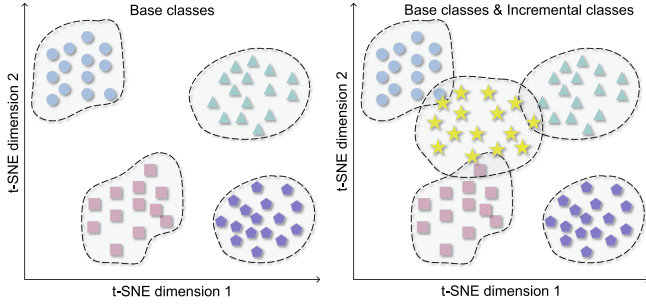


Fig. 5. Class separation and overlap in few-shot incremental learning.

Here, z_i is the feature embedding of sample i , $P(i)$ represents the set of positive samples from the same class as i , $A(i)$ denotes the set of all samples in the batch, and τ is a temperature parameter that adjusts the concentration of the similarity distribution. SupCon effectively improves feature discriminability in few-shot scenarios, enhancing incremental classification performance. Together with CosFace in the few-shot training phase, they constitute a forward-backward compatibility mechanism to address feature overlap. CosFace enforces angular margin constraints to reserve space for future classes, achieving forward compatibility. SupCon with old class exemplar replay maintains inter-class separation and intra-class compactness during incremental training, achieving backward compatibility. This dual-phase design enables accommodation of novel attacks while preserving performance on established threats.

V. SIMULATION RESULTS AND DISCUSSION

A. Dataset

As far as we know, no standardized benchmark dataset exists for incremental few-shot learning in intrusion detection [135]. For real intrusion detection scenarios, the selected datasets must have:

- 1) normal and diverse malicious traffic classes reflecting real-world network conditions;
- 2) PCAP format preserving raw network traffic bytes;
- 3) a few labeled data per traffic type to support incremental few-shot learning paradigms.

We select two open datasets, i.e., CIC-IDS2017 [136] and USTC-TFC2016 [10], to evaluate the effectiveness of our proposed method.

CICIDS2017¹ serves as an established benchmark in the field of intrusion detection, having been developed by the Canadian Institute for Cybersecurity and extensively utilized by researchers worldwide. Capturing five days of network traffic from July 3 to July 7, 2017, it encompasses both benign flows and 14 types of contemporary malicious traffic, including DDoS, Web Attacks, Botnets, etc. The dataset provides raw packet captures in PCAP format, retaining the original traffic bytes, alongside CSV files with over 80 extracted flow features generated by CICFlowMeter [137], each labeled with timestamps,

TABLE II
CICIDS2017 DATASET SESSION SETTINGS FOR INCREMENTAL FEW-SHOT LEARNING

Session 0	Session 1	Session 2
Benign, DoS-Hulk, DoS-Slowloris, DoS-SlowHTTPTest, DDoS, PortScan, DoS-GoldenEye	Bot, SSH-Patator	WebAttackxss, FTP-Patator

source/destination IPs, ports, protocols, and attack types. Due to the limited number of samples for certain attack types and labeling inconsistencies in specific instances [12], we select 10 attack types with sufficient sample sizes, employing raw traffic data in PCAP format as input for experimental evaluation of the proposed scheme. This selection enhances the dataset's suitability for evaluating our system's performance across diverse attack scenarios. For incremental few-shot learning experiments, we structured the CICIDS2017 dataset into session sets, as shown in Table II. In Session 0, consisting of benign traffic and 6 malicious traffic types, we allocated 5,000 labeled training samples per traffic type to contain sufficient data. For incremental learning, sessions 1 and 2 each includes 5 labeled samples per type to simulate few-shot learning scenarios. Performance evaluation after each session uses 500 test samples per type.

The USTC-TFC2016 dataset,² developed by researchers at the University of Science and Technology of China (USTC), represents a comprehensive resource designed for advanced research in encrypted traffic classification and intrusion detection. Captured in a real-world network environment and stored in PCAP format with a compressed size of approximately 3.71GB, it encompasses both benign and malicious traffic flows. The dataset includes 10 types of benign traffic, generated using the IXIA BPS [138] network traffic simulation tool, covering applications such as BitTorrent, Gmail, Skype, etc. Additionally, it contains 10 types of malicious traffic, collected researchers at the Czech Technical University (CTU) [139] from public websites in real network environments between 2011 and 2015, including malware such as Cridex, Geodo, Htbot, etc. This diverse dataset enables robust evaluation of intrusion detection systems under realistic conditions. For incremental few-shot learning, all benign traffic in the USTC-TFC2016 dataset is grouped into one class, combined with 6 randomly selected malicious traffic types as base classes, each with 5,000 labeled training samples. The remaining four malicious types serve as incremental classes, each with 5 training samples. During evaluation, each class, whether base or incremental, consists of 500 test samples to ensure a balanced assessment. The few shot incremental session setup for the USTC-TFC2016 dataset is shown in Table III.

B. Experiment Setting

All algorithms in this paper are implemented using PyTorch 2.2.2, with training and testing conducted on an NVIDIA GeForce RTX 4090 graphics card. The experimental setup for

¹ <https://www.unb.ca/cic/datasets/ids-2017.html>

² <https://github.com/davidyslu/USTC-TFC2016>

TABLE III
USTC-TFC2016 DATASET SESSION SETTINGS FOR INCREMENTAL FEW-SHOT LEARNING

Session 0	Session 1	Session 2
Benign, Cridex, Geodo, Zeus Miuref, Neris, Virut,	Htbot, Nsis_ay	Shifu, Tinba

TABLE IV
HYPERPARAMETER SELECTION FOR CONVOLUTIONAL NEURAL NETWORKS

Parameter	Value
Input Size	64×64
Learning Rate	0.001
Batch Size	64
Optimizer	Adam
Training Epochs	100
Dropout Rate	0.2
Weight Decay	$1e-4$
Number of Convolutional Layers	4
Kernel Size	3×3
Pooling Type and Size	Max Pooling, 2×2
Activation Function	ReLU
Fully Connected Layer Neurons	512
Loss Function	Cross-Entropy loss

our proposed method is outlined in three stages: pre-training, few-shot learning, and few-shot incremental fine-tuning.

1) *Pre-Training*: As described in Section III-B, we extract bytes from each bidirectional flow by selecting the first L bytes of the first M packets. Referring to [12], [17], we set $M = 16$ and $L = 256$ to capture critical features for traffic classification. The normalized bytes are reshaped into a 64×64 matrix for CNN model input. The CNN hyperparameters, including learning rate, batch size, and optimizer type, are configured as specified in Table IV.

2) *Few-Shot Learning*: For the few-shot learning task, we selected 7 base classes from the CICIDS2017 and USTC-TFC2016 datasets, which correspond to the data-rich known classes used in the pretraining phase, and configured the experiments as follows:

- *Episode configuration*: 7-way 5-shot with 5 support samples and 10 query samples per class, totaling 35 support and 70 query samples per episode.
- *Training*: 20 epochs with 800 episodes per epoch using Adam optimizer with learning rate scheduling.
- *Model and loss*: CNN backbone, initialized with pretrained weights, extracts 512-dimensional features for prototypical network classification. Training objective employs Cos-Face loss ($s = 30.0$, $m = 0.35$).
- *Evaluation*: 500 episodes with randomly sampled 7-way 5-shot 10-query tasks.

3) *Few-Shot Incremental Fine-Tuning*: For the few-shot incremental learning phase, new classes are introduced in each session with limited labeled samples. The experiments are configured as follows:

- *Episode configuration*: k_i -way 5-shot, where k_i represents the total number of classes from current and previous sessions. Each new class contains 5 support samples, while

5 exemplars per old class are retained to maintain balanced representation.

- *Training*: The model is fine-tuned for 30 epochs per session. The convolutional layers of the CNN are frozen while only the fully connected layer parameters undergo fine-tuning to adapt to new classes.
- *Model and loss*: The samples are processed through the CNN and prototypical network. Training objective uses supervised contrastive loss with temperature $\tau = 0.07$.
- *Evaluation*: 500 episodes with randomly sampled k_i -way 5-shot 10-query tasks across all learned classes.

C. Performance Metrics

In multi-class classification, model performance is typically assessed using macro-averaged metrics: Accuracy, Precision, Recall, F1-Score, and the confusion matrix. We define the following terms for each class j , with C representing the total number of classes: TP_j is the number of instances belonging to class j and correctly classified as class j , FP_j represents the number of instances not belonging to class j but incorrectly classified as class j , TN_j is the number of instances not belonging to class j and correctly classified as such, and FN_j is the number of instances belonging to class j but incorrectly classified as other classes. The evaluation metrics are defined as follows: Accuracy evaluates the overall correctness of the model across all classes:

$$\text{Accuracy} = \frac{\sum_{j=1}^C TP_j}{\sum_{j=1}^C (TP_j + FN_j)}. \quad (10)$$

Macro-Precision is used to evaluate the correctness of the model in predicting instances as belonging to each class:

$$\text{Macro-Precision} = \frac{1}{C} \sum_{j=1}^C \frac{TP_j}{TP_j + FP_j}. \quad (11)$$

Macro-Recall is used to evaluate the coverage of the model in identifying instances that truly belong to each class:

$$\text{Macro-Recall} = \frac{1}{C} \sum_{j=1}^C \frac{TP_j}{TP_j + FN_j}. \quad (12)$$

Macro-F1 provides a balanced evaluation of Precision and Recall across classes:

$$\text{Macro-F1} = \frac{1}{C} \sum_{j=1}^C 2 \times \frac{\text{Precision}_j \times \text{Recall}_j}{\text{Precision}_j + \text{Recall}_j}. \quad (13)$$

The confusion matrix evaluates classification errors of a model by comparing predicted class labels with actual ones.

Inspired by [112], we adopt harmonic accuracy to address the shortcomings of average accuracy in few-shot incremental learning. Average accuracy may be biased by strong performance on base classes, thereby obscuring inadequate learning of new classes. For example, a model achieving 100% accuracy on 20 base classes (each with 5 test samples) but 0% on 2 new classes (each with 5 test samples) results in a misleading average

TABLE V
ABLATION STUDY RESULTS ON CIC-IDS2017 AND USTC-TF2016 DATASETS

Components			CIC-IDS2017				USTC-TF2016			
Pretraining	Few-shot learning	Few-shot incremental	Acc. in each session (%)			PD ↓	Acc. in each session (%)			PD ↓
			0	1	2		0	1	2	
×	✓	×	99.78	92.10	90.06	9.72	97.42	82.64	76.71	20.71
✓	✓	×	99.84	96.43	94.00	5.84	98.20	85.16	80.01	18.19
✓	✓	✓	99.86	99.04	98.64	1.22	98.31	92.04	90.32	7.99

accuracy of 90.9% . Harmonic accuracy, defined as

$$A_h = \frac{2 \times A_b \times A_i}{A_b + A_i}, \quad (14)$$

where A_b represents the average accuracy on base classes and A_i represents the average accuracy on incremental classes, ensures balanced performance across all classes. A high harmonic accuracy reflects robust learning on both base and new classes, while a low value indicates deficiencies in incremental learning capability.

D. Experiment Results and Analysis

1) *Ablation Analysis*: To validate the effectiveness of each component, we conduct an ablation study, systematically evaluating the impact of individual modules on overall performance, with results presented in Table V . Specifically, on the CIC-IDS2017 dataset, training solely with base classes results in a performance degradation (PD) as high as 9.72% after two sessions of incremental learning. At this stage, the model is not exposed to new category samples, leading to a low recognition rate for new categories. Introducing pretraining reduces PD to 5.84% , demonstrating its ability to enable the model to acquire robust knowledge, effectively learning old categories while enhancing adaptability to new tasks. Furthermore, incorporating few-shot incremental fine-tuning substantially enhances model effectiveness. By freezing the backbone network and fine-tuning the fully connected layers, the model effectively learns the knowledge of new category samples while retaining the knowledge of old categories. Consequently, on the CIC-IDS2017 dataset, the PD decreases to 1.22% , with accuracies across sessions 0 to 2 reaching 99.86% , 99.04% , and 98.64% , respectively. Similarly, on the USTC-TF2016 dataset, training with only few-shot base classes results in a PD of 20.71% , but with all components, PD drops to 7.99% , further validating our proposed three-stage few-shot incremental learning intrusion detection system. Notably, the USTC-TF2016 dataset shows reduced detection accuracy compared to CIC-IDS2017, probably because it provides complete PCAP files for each traffic category over a period. Although primarily aligned with their labels, these files contain mixed traffic types which cause labeling errors in bidirectional flows. In contrast, CIC-IDS2017 utilizes manual label calibration with IP and time segment data [112], ensuring higher accuracy. Despite such label noise, our model sustains optimal classification performance, highlighting its robustness and adaptability.

To demonstrate the advantages of our proposed approach in online intrusion detection intuitively, we employed t-SNE [140] to visualize the feature embeddings of test samples in a low-dimensional space, as shown in Fig. 6. In the session 0 base class test shown in Fig. 6(a), few-shot training using the CosFace loss function achieves significant inter-class separation for the feature embeddings of 7 base classes, preserving sufficient feature space for novel class embeddings in subsequent incremental learning. In the session 1 incremental test shown in Fig. 6(b) (7 base classes and 2 novel classes) and the session 2 incremental test illustrated in Fig. 6(c) (7 base classes and 4 novel classes), incremental fine-tuning with the SupCon loss function, combined with the Exemplars replay strategy for base classes, ensures complete separation between base and novel class embeddings in the feature space. Consequently, when novel attack classes emerge, the proposed detection strategy efficiently learns novel class features while retaining base class feature memory after a few fine-tuning epochs, thereby achieving superior detection performance.

To illustrate the effectiveness of our proposed online intrusion detection framework, we generated a confusion matrix, as shown in Fig. 7, using 500 randomly sampled episodes. In Session 0, the base class classification, illustrated in Fig. 7(a), achieves superior accuracy. As shown in Fig. 7(b), our model additionally detects two novel attack classes, BoT and SSH-Patator, with accuracies of 95% and 99% , respectively. It also sustains base class detection performance with negligible impact. In Session 2, subsequent incremental fine-tuning enables the seamless integration of two additional classes, WebAttackXSS and FTP-Patator, as shown in Fig. 7(c), achieving classification accuracies of 92% and 100% , respectively, while maintaining robust categorization performance for the base classes and previously introduced novel classes. These results demonstrate the effectiveness of the proposed scheme for IDS with limited samples.

2) *Comparison With Different Methods*: In cybersecurity, research on few-shot incremental learning for intrusion detection is relatively limited. To ensure a comprehensive and fair comparison, we include representative methods from both existing few-shot class-incremental learning approaches for intrusion detection and conventional non-incremental baselines, along with several typical FSCIL techniques adapted from the computer vision domain. Wang et al. [16] proposes a meta-learning few-shot incremental learning strategy employing network traffic features rather than raw bytes for classification. Di Monda et al. [18] established a paradigm for few-shot incremental learning in intrusion detection, evaluating several common methods on intrusion

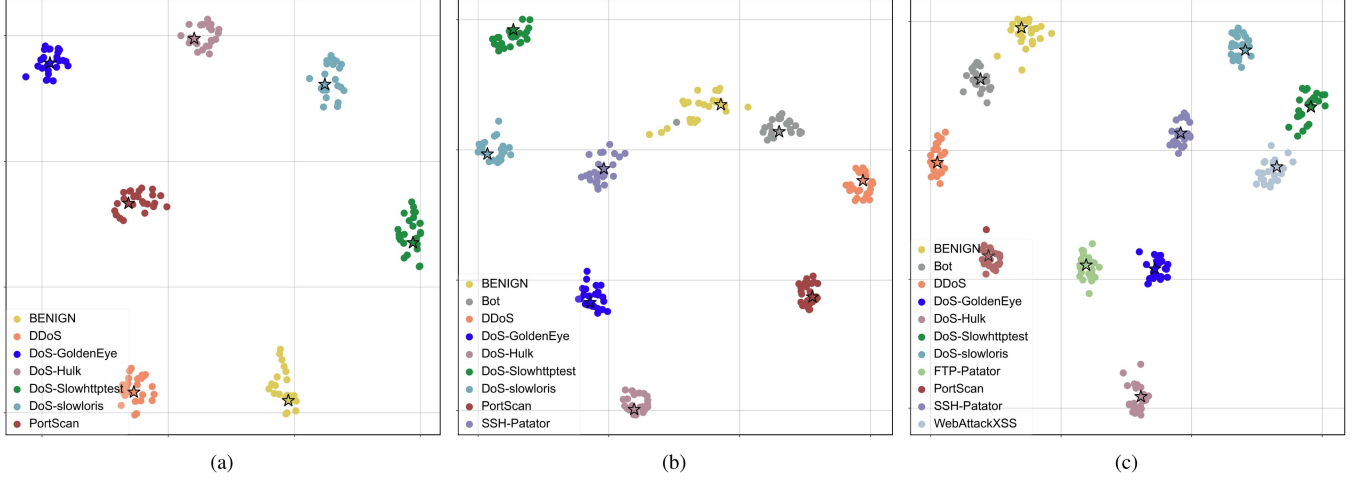


Fig. 6. t-SNE visualization of class increment in few-shot learning (CICIDS2017 dataset): (a) Base classes; (b) After session 1 with 2 added classes; (c) After session 2 with 2 more classes (4 new classes).

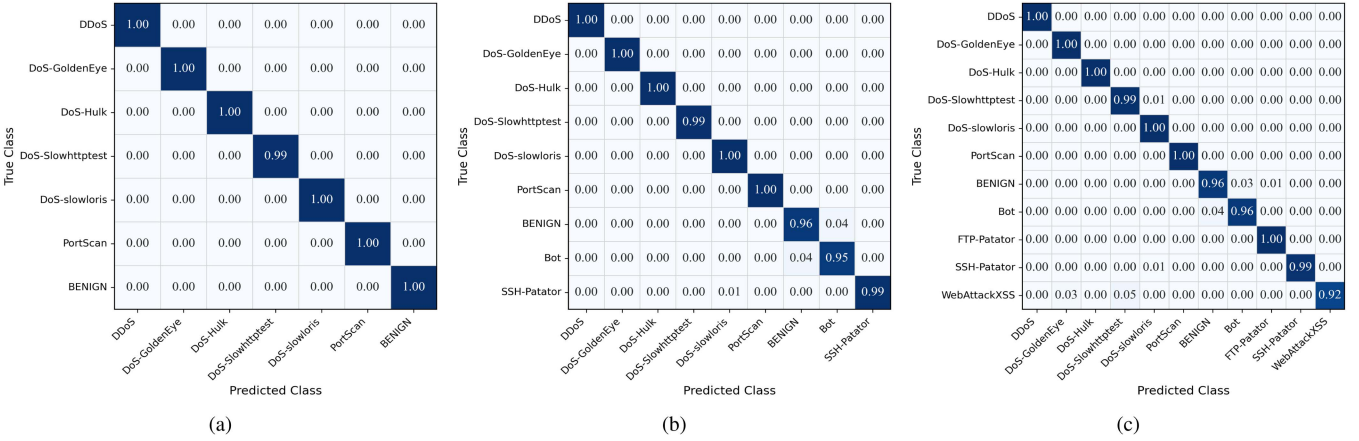


Fig. 7. Confusion matrix of class increment in few-shot learning (CICIDS2017 dataset): (a) Base classes; (b) After session 1 with 2 added classes; (c) After session 2 with 2 more classes (4 new classes).

TABLE VI
COMPARISON OF DIFFERENT METHODS ON CIC-IDS2017 AND USTC-TFC2016 DATASETS

Method	CIC-IDS2017				USTC-TF2016			
	Acc. in each session (%)			PD ↓	Acc. in each session (%)			PD ↓
	0	1	2		0	1	2	
Non-few-shot [10]	99.86	95.67	94.27	5.59	98.14	82.33	81.45	16.69
Finetune	99.89	78.00	75.73	24.16	98.26	78.93	64.55	33.71
SoftNet [117]	99.65	88.38	88.04	11.61	96.96	74.61	72.30	24.66
CLOSER [114]	99.85	89.46	87.82	12.03	98.06	82.80	73.47	24.59
Prototype Network [13]	99.80	90.11	87.80	12.00	96.76	78.75	68.67	28.09
Pre-train + Prototypical Classifier [141]	99.88	93.08	90.39	9.49	98.24	81.62	71.56	26.68
ALICE [112]	99.72	91.10	90.05	9.67	96.84	78.43	74.60	22.24
Ours	99.86	99.04	98.64	1.22	98.31	92.04	90.32	7.99

detection datasets. We systematically compare it with traditional deep learning methods [10], pre-trained and fine-tuned method, prototype-based few-shot learning approaches [13], and recent few-shot incremental learning techniques adapted from image processing [112], [114], [117], [141] to enhance the comparative

framework, as shown in Table VI. For a consistent and fair evaluation, all methods adopt the data processing approach from Session III-B, which segments traffic into sessions and uses the first 256 bytes of the first 16 packets per session as the raw byte features of the traffic. Specifically:

- *Non-few-shot* [10] divides the labeled base-class traffic with sufficient samples into training, validation, and test sets at a 6:2:2 ratio, resizing the 16×256 raw byte features into 64×64 grayscale images for supervised CNN classification. When new class samples emerge, the method combines existing base-class data with the new class samples for CNN retraining.
- *Finetune* adopts the same base-class training procedure as non-few-shot. Upon the arrival of new-class samples, the approach freezes the backbone network, fine-tuning only the fully connected layer with these new class samples.
- *SoftNet* [117] proposes an adaptive learning framework, partitioning parameters into primary and secondary subnetworks, designating the top $c\%$ (here, 80%) critical parameters as the primary subnetwork during base-class training, with the remainder as the secondary subnetwork, and updating only the secondary subnetwork in the incremental phase to adapt to new class samples while mitigating forgetting of prior class knowledge.
- *CLOSER* [114] introduces a class compaction loss to enhance representation learning. It optimizes feature space compactness and transferability by compressing inter-class distances, facilitating efficient acquisition of new class features. Meanwhile, it improves class prototype discriminability and base-class performance by compressing intra-class distances.
- *Prototype Network* [13] applies few-shot learning with prototypical networks to intrusion detection, constructing multiple k -way n -shot q -query tasks from base-class data for feature extraction and training with the prototypical network as the classifier. In the incremental phase, it uses the optimized feature extraction network and prototypical network from base-class training to classify samples.
- *Pre-train + Prototypical Classifier* [141] proposes integrating pre-trained feature embeddings with a prototypical network classifier, using a pre-trained model to extract universal features, applying L2 normalization to standardize feature norms and reduce intra-class variance, and optimizing feature distribution via embedding space transformation for better transferability, enabling few-shot incremental learning with the prototypical network classifier.
- *ALICE* [112] introduces an angular penalty loss instead of cross-entropy loss to enhance intra-class feature compactness and inter-class feature separability, thereby facilitating integration of new classes in incremental learning. It also employs class and data augmentation through pseudo-class generation to improve model generalization and enhance the feature extractor's ability to adapt to new categories.

As shown in Table VI, our method outperforms other approaches in terms of accuracy during incremental sessions, while maintaining the lowest PD value. This performance benefits from angular margin-based loss that enlarges inter-class distances during base training and contrastive exemplar replay that maintains class separability during incremental learning, reducing feature interference while preserving prior knowledge.

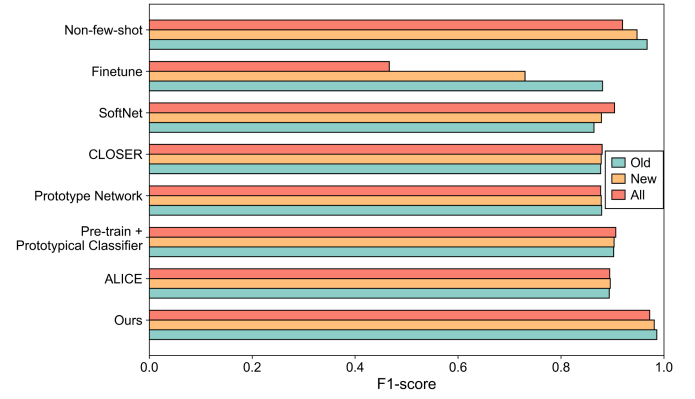


Fig. 8. Performance comparison with old, new, and all categories on the CIC-IDS2017 dataset.

Compared to traditional non-few-shot methods [10], which require extensive retraining with base and new class attack data, incurring high latency and substantial data storage, our approach reduces PD by 4.37% and 8.7% on the CIC-IDS2017 and USTC-TFC2016 datasets, respectively, while adapting to new attacks with minimal fine-tuning epochs and few stored old class exemplars, effectively maintaining performance on old classes. Fine-tuning in few-shot incremental settings underperforms due to insufficient new samples and degraded performance on old classes. Compared to fine-tuning with frozen network layers, SoftNet [117] fine-tuning secondary parameters in a subnet and replaying old class exemplars, achieving commendable performance in both accuracy and PD metrics. However, its capacity to balance adaptation and retention remains limited due to its constrained update flexibility. CLOSER [114] and ALICE [112] utilize refined forward-compatible feature embeddings, performing robustly on the CIC-IDS2017 dataset but inadequately on USTC-TFC2016, indicating limited robustness to dataset-specific variations. Prototype Network [13] and Pre-train + Prototypical Classifier [141] employ meta-learning-based prototypical networks for effective classification. Through pre-training, Pre-train + Prototypical Classifier [141] significantly improves feature extraction, facilitating effective incremental learning. Nonetheless, both methods lack an incremental adaptation process, limiting their ability to effectively model new classes under distributional shifts.

Figs. 8 and 9 performance the evaluation of different methods for old, new, and all categories on the CIC-IDS2017 and USTC-TFC2016 datasets. Our method achieves the highest F1-scores on the CIC-IDS2017 dataset with 0.9862 for old, 0.9725 for new, and 0.9812 for all categories. It also achieves 0.9347 for old, 0.8503 for new, and 0.9040 for all categories on the USTC-TFC2016 dataset. These results demonstrate strong generalization to both historical and emerging attacks across datasets with varying traffic characteristics. The non-few-shot [10] method achieves a high F1-score for old categories and a low F1-score for new categories. Such performance results from incremental retraining with abundant old class samples and only 5 new class samples, leading to class imbalance that restricts new class detection. Fine-tuning-based methods exhibit reduced accuracy

TABLE VII
ANALYSIS OF DIFFERENT METHODS USING PRECISION, RECALL, AND HARMONIC ACCURACY METRICS (AFTER SESSION 2)

Method	CIC-IDS2017			USTC-TF2016		
	Precision	Recall	Harmonic Accuracy	Precision	Recall	Harmonic Accuracy
Non-few-shot [10]	0.9656	0.9473	0.9218	0.8587	0.8145	0.6711
Finetune	0.7163	0.7787	0.7573	0.6820	0.6455	0.1175
SoftNet [117]	0.8820	0.8795	0.8919	0.7226	0.7273	0.6651
CLOSER [114]	0.8783	0.8782	0.8777	0.7356	0.7363	0.7383
Prototype Network [13]	0.8788	0.8780	0.8792	0.6925	0.6867	0.6871
Pre-train + Prototypical Classifier [141]	0.9033	0.9033	0.9035	0.7182	0.7175	0.7162
ALICE [112]	0.9035	0.9005	0.9065	0.7574	0.7460	0.7347
Ours	0.9780	0.9818	0.9813	0.9182	0.9034	0.8927

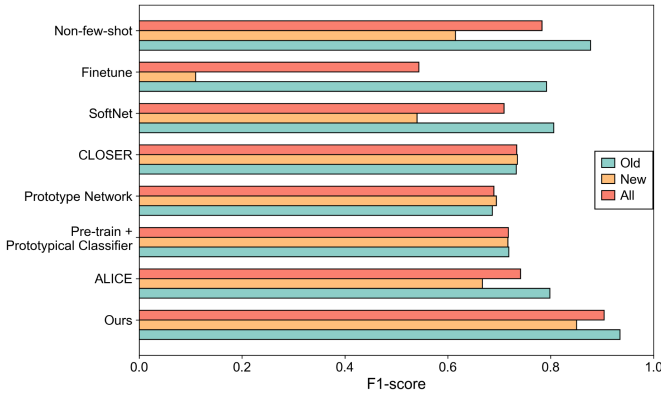


Fig. 9. Performance comparison with old, new, and all categories on the USTC-TF2016 dataset.

for old classes due to parameter adjustment and low accuracy for new classes, as limited samples impede effective learning of new class features during optimization. ALICE [112] performs lower F1-scores for new categories compared to old categories on the USTC-TFC2016 dataset, alongside diminished old class accuracy. Such performance arises from feature space overlap between new and old classes, since the method lacks further adaptation of the feature space during the incremental stage. SoftNet [117] also struggles with insufficient new class learning on the USTC-TFC2016 dataset, where new and old classes exhibit substantial distributional differences. Its constrained sub-net fine-tuning, while preserving old class knowledge, proves inadequate for learning discriminative features under such distributional shifts. Other methods display balanced performance across new and old classes but remain less competitive than our method overall. Additionally, Table VII presents the analysis of different methods using macro-average Precision, Recall, and Harmonic Accuracy metrics, demonstrating that our method achieves the highest performance across all metrics on both CIC-IDS2017 and USTC-TFC2016 datasets, achieving optimal online intrusion detection.

3) *Complexity Analysis*: We analyze the computational complexity of the proposed few-shot class-incremental learning framework. The system undergoes two-stage training. Pre-training the CNN backbone with 2.49M parameters on base classes requires 100 epochs with 28000 samples and 438 iterations per epoch. With computational cost $F_{\text{iter}} = 3 \times F_{\text{forward}}$

where $F_{\text{forward}} = 120.73$ MFLOPs, as the backward pass is approximately twice the cost of the forward pass, the total pre-training complexity is $F_{\text{pretrain}} \approx 1014$ TFLOPs. Meta-training employs 16,000 episodic tasks at 800 tasks per epoch over 20 epochs. Each 7-way 5-shot task contains 105 samples (35 support, 70 query), yielding per-task complexity $F_{\text{task}} \approx 29.58$ GFLOPs and total meta-training cost $F_{\text{meta}} \approx 473.34$ TFLOPs. Although total offline training requires 1487.34 TFLOPs, it is performed once during the training phase, producing an INT8-quantized model of only 2.37 MB for practical deployment.

For edge inference, each sample requires 120.73 MFLOPs. On commonly used edge devices such as NVIDIA Jetson Xavier NX (21 TOPS INT8) deployed in edge networks, inference latency reaches 0.006 ms, satisfying real-time requirements. For incremental learning with 5 new attack samples, the model fine-tunes over 20 epochs using 40 samples per epoch, requiring $F_{\text{incr}} \approx 289.75$ GFLOPs. Employing FP16 mixed-precision training strategy, model updates can be completed in approximately 50–60 seconds on edge devices. With 2.49M parameters, 120.73 FLOPs per inference, and 2.37 MB footprint, the framework enables efficient edge deployment with rapid adaptation to evolving threats.

The experimental results provide critical insights for deploying adaptive intrusion detection in edge networks. Specifically, our experiments show that operators can perform online model updates with just 5–10 high-quality samples per new attack class, significantly reducing annotation burden while enabling rapid adaptation to evolving threats.

VI. CONCLUSION

In this paper, we propose a multi-stage few-shot class-incremental learning framework for edge network intrusion detection. It integrates pre-training, CosFace-based training, and SupCon-based contrastive learning to enable rapid new class adaptation and mitigate old class forgetting in few-shot scenarios. Compared to existing methods, our approach delivers superior incremental detection performance on the CIC-IDS2017 and USTC-TFC2016 datasets. However, the reliance of our scheme on known samples limits the ability to detect unseen attacks. In future work, we will explore an efficient FSCIL model for generalized open-set and zero-shot intrusion detection of novel attacks, as well as addressing imbalanced sample distributions in edge networks.

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Zhen Yin is currently working toward the post-graduation degree with the College of Electronic and Information Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China. Her research interests include wireless security, intrusion detection, deep learning, and meta-learning.



Wei Wang (Member, IEEE) received the B.Eng. degree in information countermeasure technology and the M.Eng. degree in signal and information processing from Xidian University, Xi'an, China, in 2011 and 2014, respectively, and the Ph.D. degree in electrical and electronic engineering from Nanyang Technological University, Singapore, in 2018. He was a Professor with the Nanjing University of Aeronautics and Astronautics, from 2019 to 2025, and a Postdoctoral Fellow with the Department of Electrical and Computer Engineering, University of Waterloo,

Waterloo, ON, Canada, from 2018 to 2019. He is currently a Professor with the School of Information and Communication Engineering, Xi'an Jiaotong University, Xi'an. His research interests include wireless communications, space-air-ground integrated networks, and wireless security. He was awarded the Chinese government award for outstanding self-financed students abroad in 2018, and the Young Elite Scientist Sponsorship Program, China Association for Science and Technology in 2021.



Yanyu Cheng (Senior Member, IEEE) received the B.S.Eng. degree in information engineering from Shanghai Jiao Tong University, Shanghai, China, in 2015, and the M.Sc. degree in signal processing and the Ph.D. degree in electrical and electronic engineering from Nanyang Technological University (NTU), Singapore, in 2016 and 2021, respectively. From 2021 to 2023, he was a Research Fellow with Alibaba-NTU Singapore Joint Research Institute, NTU. He is currently a Visiting Professor with the Joint NTU-UBC Research Centre of Excellence in Active Living for

the Elderly (LILY), NTU, and a Tenure-Track Professor with the School of Cyberspace, Hangzhou Dianzi University, Hangzhou, China. His research interests include networks, cybersecurity, and AI-enabled intelligent systems.



Yiliang Liu (Member, IEEE) received the B.E. and M.Sc. degrees in computer science and communication engineering from Jiangsu University, Zhenjiang, China, in 2012 and 2015, respectively, and the Ph.D. degree from the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin, China, in 2020. He is currently an Associate Professor with the School of Cyber Science and Engineering, Xi'an Jiaotong University, Xi'an, China. His research interests include wireless communication security, physical-layer security, and intelligent connected vehicles. He was the recipient of the Outstanding Doctoral Dissertation Award from China Education Society of Electronics in 2020 and the Best Paper Award from the IEEE Systems Journal in 2021, IEEE VTC-Fall in 2023, IEEE ICITE in 2024, and IEEE AIoT in 2024.



Xiaozhen Lu (Member, IEEE) received the B.S. degree in communication engineering from the Nanjing University of Posts and Telecommunications, Nanjing, China, in 2017, and the Ph.D. degree in communication and information systems from Xiamen University, Xiamen, China, in 2021. She is currently an Associate Professor with the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing. Her research interests include reinforcement learning, federated learning, network security, and wireless communications. She was the recipient of the Best Student Paper Award for ML4CS 2019 and the Excellent Paper Award for CWSN 2020.